

Activity 4: Generating a sequence of all fractions between 0 and 1

CHALLENGE

Can we generate all the *proper fractions*?

Task A: First generate all proper fractions whose denominator is 100.

A *proper fraction* is a fraction where the number on top is less than the number below.



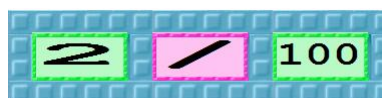
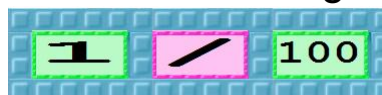
The *denominator* of a fraction is the number underneath. In this task we are generating all the hundredths from $1/100$ to $99/100$.



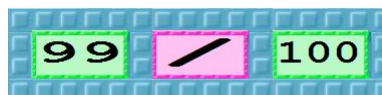
When given the box



the Next Numerator robot should give the bird 99 boxes:



...



The robot can be trained to give the bird one box and will stop when the scale stops tilting towards the number 100.

If you need help training Next Numerator ask for pictorial instructions.

Task A: First generate all proper fractions whose *denominator* is 100.

Task B: Use the Next Numerator robot to make other sequences of fractions.

Predict. What sequence does Next Numerator produce when given



Write your answer here before running it.

Test. Was the result what you expected?

What sequence is produced when you change the number in the Denominator box to some other number, say 89?

Generalise. Explain in general what Next Numerator does.

Task C: Generate all *denominators*

Train a robot named `Next Denominator` that will change the box so that `Next Numerator` will work on the next denominator when finished. If you get stuck ask for the pictorial instructions.

*Put `Next Denominator` **and** `Next Numerator` together to form a team. Package up your team in a box like this*



and save your work.

If the `All Fractions` team of robots ran forever would the bird ever be given two identical boxes (the same numbers as the other one)?

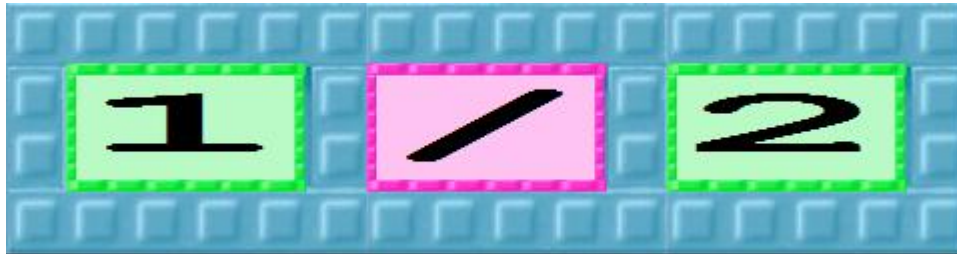
Explain.

Do you think all the *proper fractions* will be generated if the robots ran forever?

If so, how would you convince a friend?

If not, can you give a *proper fraction* that will not be generated?

Are there boxes that will be generated that are equivalent to this box?



If so, give a few examples.

Task D: No repeated fractions

Load the `No Copies` team into ToonTalk. (It can be found in the Infinity section of the Tools section of the WebLabs web site.)
Connect it to `All Fractions` with a bird and nest.

Can you describe how the `No Copies` team works?

Task E: Dancing with the fractions

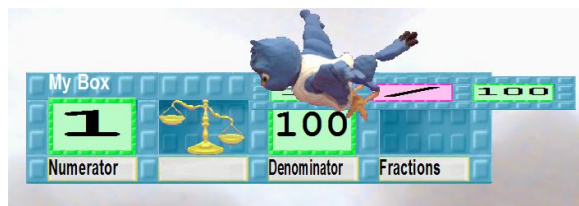
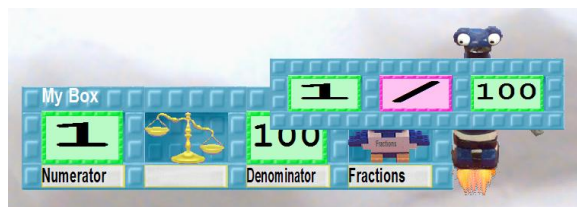
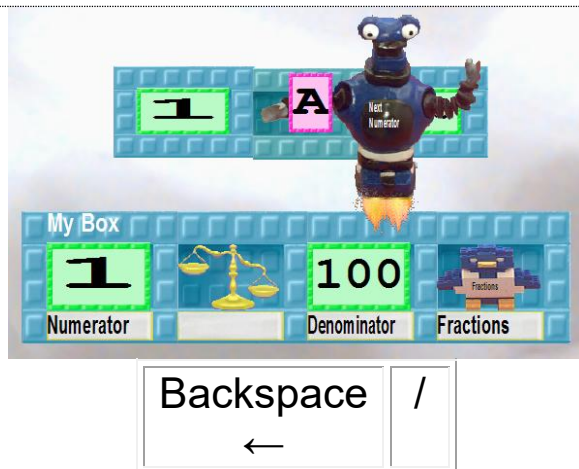
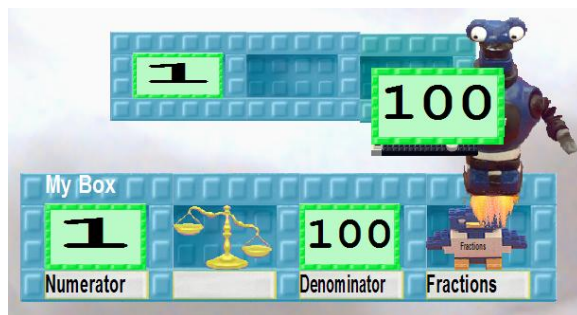
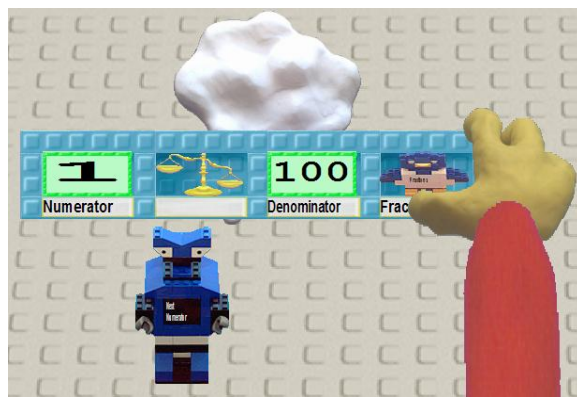
Connect the output from the `No Copies` team to the `Match Maker` robot from Activity 3.

Do you think all the fractions between 0 and 1 are countable?

Write down how you would convince a younger child of your answer.

What do you think it means for something to be *countable*?

Activity 4, Task B: Training Next Numerator



Activity 4, Task C: Training Next Denominator

